Transformation of a Continuous Random Variable

Problem Setup

Let X be a continuous random variable with PDF $f_X(x)$.

Define a new random variable

$$Y = g(X).$$

The goal is to find the PDF of Y, denoted by

$$f_Y(y).$$

Notation

The CDF of X is

$$F_X(x) = P(X \leq x).$$

The PDF of X is

$$f_X(x) = \frac{d}{dx} F_X(x).$$

Similarly, the CDF of Y is

$$F_Y(y) = P(Y \leq y),$$

and its PDF is

$$f_Y(y) = \frac{d}{dy} F_Y(y).$$

Normal Distribution

If

$$X \sim \mathcal{N}(\mu, \sigma^2),$$

then the PDF of X is

$$\boxed{f_X(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right), \qquad -\infty < x < \infty.}$$

Here,

$$E[X] = \mu,$$

and

$$\mathrm{Var}(X) = \sigma^2.$$

General CDF Method

Suppose

$$Y = g(X).$$

To find the PDF of Y:

1. Find the possible values of Y.
2. Write the CDF of Y:

$$F_Y(y) = P(Y \leq y).$$

3. Substitute $Y = g(X)$:

$$F_Y(y) = P(g(X) \leq y).$$

4. Rewrite the event $g(X) \leq y$ as a condition on X.
5. Use the CDF or PDF of X to compute $F_Y(y)$.
6. Differentiate:

$$\boxed{f_Y(y) = \frac{d}{dy} F_Y(y).}$$

Example

Let

$$X \sim \mathcal{N}(\mu, \sigma^2)$$

and define

$$Y = 2X + 3.$$

We want to find the distribution of Y.
Start with

$$F_Y(y) = P(Y \leq y).$$

Substituting $Y = 2X + 3$,

$$F_Y(y) = P(2X + 3 \leq y).$$

Since

$$2X + 3 \leq y \iff X \leq \frac{y-3}{2},$$

we get

$$F_Y(y) = F_X\left(\frac{y-3}{2}\right).$$

Differentiate with respect to y:

$$f_Y(y) = \frac{1}{2} f_X\left(\frac{y-3}{2}\right).$$

Substituting the normal PDF,

$$f_Y(y) = \frac{1}{2\sigma\sqrt{2\pi}} \exp\left(-\frac{\left(\frac{y-3}{2}-\mu\right)^2}{2\sigma^2}\right).$$

Equivalently,

$$\boxed{Y \sim \mathcal{N}(2\mu+3,\, 4\sigma^2).}$$

Hence,

$$\boxed{f_Y(y) = \frac{1}{2\sigma\sqrt{2\pi}} \exp\left(-\frac{(y-(2\mu+3))^2}{8\sigma^2}\right).}$$

Main Idea

$$\boxed{Y = g(X) \;\longrightarrow\; F_Y(y) = P(g(X) \le y) \;\longrightarrow\; f_Y(y) = \frac{d}{dy} F_Y(y).}$$